MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GLOW MODULATOR

TYPE 1B59 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Cold cathode

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ionization voltage	Operating current
Unit:	Vdc	mAdc
Maximum:	225	35 (avg) to 75 (peak)
Minimum:	---	5 average
TEST CONDITIONS:	---	---

GENERAL:

(F) Qualification - Not required

1/ Formerly tube type 1B59/R1130B

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
---	Glass and spot check	Ebb/Ib = 30 mAdc (see note 1)	---	---	---	---	---	---
	Quality conformance inspection, part 2							
1031	Vibration	t = 30; no voltages	---	---	---	---	---	---
1031	Vibration	t = 300; no voltages	---	---	---	---	---	---
3347	Ionization	Rp = 4,000 ohms	---	---	Ez	---	225	Vdc
3337	Voltage drop	Ebb/Ib = 30 mAdc	---	---	Etd	---	150	Vdc
---	Light output and contrast	Ebb/Ib = 40 mAdc	---	---	Ib(1)	0.30	---	μAdc
		Ebb/Ib = 30 mAdc	---	---	Ib(1)	48	68%Ib(1)	μAdc
		Ebb/Ib = 20 mAdc	---	---	Ib(1)	20	28%Ib(1)	μAdc
		Ebb/Ib = 10 mAdc (see note 2)	---	---	Ib(1)	5.5	8%Ib(1)	μAdc
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; Ebb/Ib = 30 mAdc; Rp = 4,000 ohms; t = 250 hours	---	---	---	---	---	---
---	Life-test end points	Ionization	---	---	Ez	---	225	Vdc
		Voltage drop	---	---	Etd	---	165	Vdc
(F)	Periodic-check tests (see note 3)							
1026	Temperature range, operating	Ionization	---	---	Ez	---	225	Vdc
1226	Envelope temperature	Ionization; T = 70°C	---	---	Ez	---	225	Vdc

NOTES:

1. With tube inserted in gage, per Drawing 221-JAN and rotated about its main axis, glass shall be optically clear and the center of the spot shall be within the circle of the decal when viewed directly through the 3/16-inch (4.76 mm) hole.

2. With the axis of the tube perpendicular to and centered with respect to the cathode of a 929 phototube, and with a Wratten C5 filter, or equivalent, inserted between the tubes, and with the distance between the glass envelopes of the tubes 9/16 inch (14.29 mm). The sensitivity of the 929 phototube used shall be 1.70 μA when the illumination on the photocathode from a lamp operated at a color temperature of 2,870°K is 8.0 footcandles. A 929 phototube of different sensitivity under the above conditions may be used if results are corrected for 1.70 μA. Ib(1) is the phototube current from the incident light of the 1B59 tube at 40 mA for 3 minutes. There shall be no wait at 30, 20, and 10 mA.

3. If these tests have previously been performed by the tube manufacturer, and if the material, manufacturing process, and test requirements are the same as when the tests were performed, the subject test(s) need not be performed.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11, 17
DSA - ES

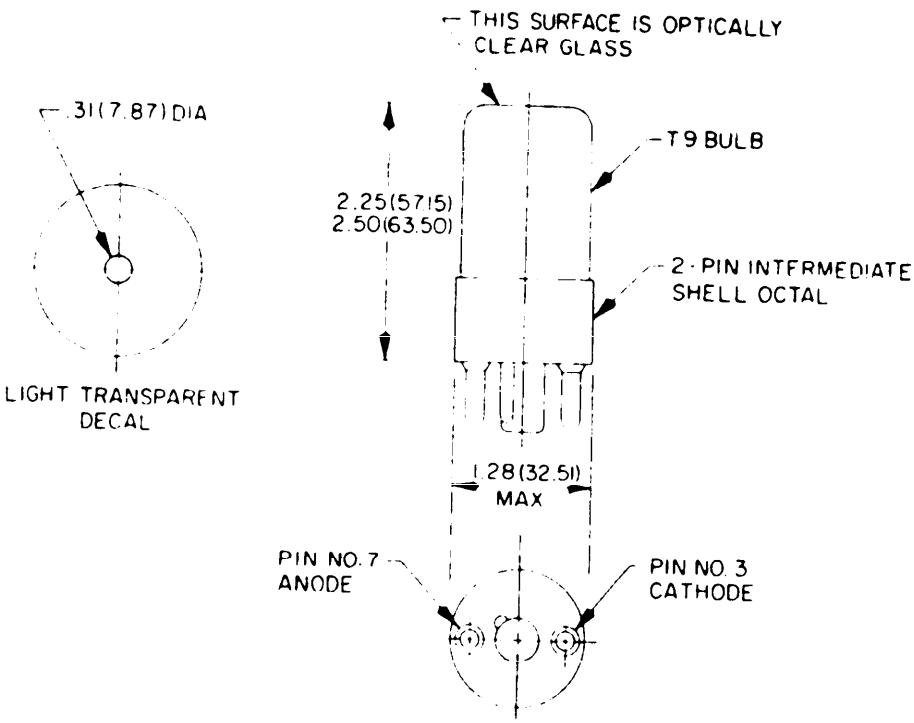
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2452)

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NOTE:

Metric equivalents are in parentheses.

FIGURE 1. Outline drawing of electron tube type 1B59.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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